protopus

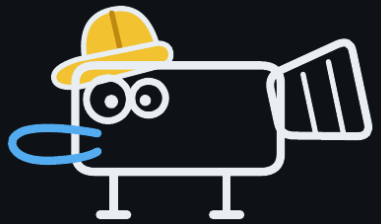
CIC-TBI · QUARTERLY PROGRESS REVIEW

Quarterly progress, July to September.

Research → design → build → integration, and what comes next.

website · Baker · Mixer · Probe Matrix

Soham, Founder · Protopus



01 · WHAT WE ARE BUILDING

Smart prototyping machines, guided by AI

TODAY fab order → 7–20 days + customs → hand soldering → a borrowed bench

PROTO

Proto

the AI engineer: plans the build, checks the design, explains results

STEP 1 · NOW

Baker

makes a finished board on the desk: mill, mask, stencil, paste, reflow

STEP 2

Mixer

tests the board: eight instruments on one cable

CONCEPT

Probe Matrix

spring probes under the board, so Proto reaches every net

FOR PROTOTYPING the first 1–10 boards of a design. Volume stays with the fabs.

Research → design → build → integration

MILESTONE	MONTH	STATUS
Market, user and competition research	Jul	DONE
Brand identity and website live	Jul	DONE
Samuel Beek (Schematik): funding offered after the proof of concept	Jul	DONE
Prototype design finalised	Aug	DONE
Build of the first Baker prototype	Aug–	IN PROGRESS
Software–hardware integration	Sep–	IN PROGRESS
Proof-of-concept funding from STMicroelectronics	Sep–	IN DISCUSSION
Probe Matrix accessory	Sep	CONCEPT
Cost report, workflow, China plan, pitch	Sep	DONE



03 · JULY 2026

Research, brand and website, first investor

1.1 Problem study — DONE

a first board: fab order, 7–20 days, ~₹6,700 per assembled try landed in India

1.2 User study — DONE

eight rungs from idea to iteration; rungs 5–8 are physical, where we work

1.3 Competition study — DONE

mills, ink printers, pick-and-place, USB instruments, fabs, AI tools: each solves one step

1.4 Market study — DONE

5,868 AICTE colleges, ~2,250 hardware startups, 10,000 Atal Tinkering Labs

1.5 Product concept — DONE

Proto plans, Baker makes, Mixer tests; for the first 1–10 boards of a design

1.6 Brand and website — DONE

name, logo, mascot, design language; site live with Baker and Mixer pages

1.7 Meeting with Samuel Beek, Schematik — DONE

verbal commitment to fund Protopus after the proof of concept

04 · AUGUST 2026

Prototype design finalised; build started

2.1 Baker 15-step process — DONE

feed, probe, mill, drill, mask, stencil, paste, jig, reflow, test

2.2 Machine architecture — DONE

one shuttle, two zones, one coordinate frame; refill magazine

2.3 Subsystem specification — DONE

500 W spindle, 10 W laser, 405 nm LCD, 138 °C reflow; Pi 5 + Klipper

2.4 Mixer architecture — DONE

eight instruments on one FPGA; for after the Baker proof of concept

2.5 Parts and bill of materials — DONE

every subsystem selected, so procurement could start

2.6 Build of the real prototype — IN PROGRESS

procurement and assembly of the first Baker



Baker, render of the design

05 · SEPTEMBER 2026 · PART 1

Software meets hardware; the business on paper

3.1 Baker Studio and Proto — IN PROGRESS

reads designs (Flux.ai API), flags problems, plans the job

3.2 Software–hardware integration — IN PROGRESS

Baker Studio to the control electronics through Klipper

3.3 Costed BOM and cost report — DONE

70 lines, costed with duty and a 15% contingency

3.4 Workflow document — DONE

“the story of a PCB”: one board through all 15 steps

3.5 China manufacturing plan — DONE

turnkey in China; testing, minimal assembly and shipping from India

₹3,73,418

Baker prototype build

₹1,44,842

Baker production (projected)

₹74,847

Mixer prototype build

₹13,740

Mixer production, projected

06 · SEPTEMBER 2026 · PART 2

Funding talks, a new accessory, a sharper plan

3.6 · IN DISCUSSION

STMicroelectronics

We approached STMicroelectronics to fund the prototype and the proof of concept. Discussions are ongoing; nothing is committed yet.

3.7 · CONCEPT

Probe Matrix

Conceived after that meeting: spring probes under the board, so Proto can reach every net to test and program it. Concept page live.

3.8 · DONE

Pitch and round-1 review

Pitch prepared; round-1 questions answered in writing. The plan is now Baker first, with funding in milestone-based instalments.

Probe Matrix concept: probe-matrix.vercel.app

07 · THE QUARTER IN NUMBERS

What the three months produced

15

Baker process steps designed

70

line items in the costed bill of materials

2 + 1

machines designed, plus one accessory conceived

₹3,73,418

costed first Baker prototype

₹1,44,842

projected Baker production cost

7

deliverables: website, BOM, cost report, workflow, China plan, pitch, concept page

08 · NEXT QUARTER · OCT–DEC 2026

Finish Baker, prove it, fund it

PLANNED STEP	TARGET
Complete the first Baker prototype build	Oct–Nov
Run all 15 Baker steps end to end on one board	Nov–Dec
Proto: plain-English brief → Gerbers → Baker job	Nov–Dec
Close proof-of-concept funding	Oct–Dec

₹15 LAKH, NINE MONTHS, RELEASED IN INSTALMENTS · NO FOUNDER SALARY

₹6L
at signing

→

₹5L
after milestone 1: Baker runs all 15 steps

→

₹4L
after milestone 2: process tests pass; Proto makes a Baker job

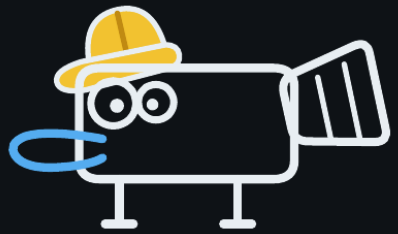
Milestone 3, the proof of concept: 20 boards made with Proto + Baker.

protopus

Thank you, sir.

Questions and guidance welcome.

WEBSITE	https://protopus-kappa.vercel.app/
BAKER	https://protopus-kappa.vercel.app/#/baker
MIXER	https://protopus-kappa.vercel.app/#/mixer
PROBE MATRIX	https://probe-matrix.vercel.app/



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